GETTING STARTED WITH MALAHIT-R1

Malahit-R1 is a portable SDR receiver that uses your mobile device, like a smartphone, for display and controls. The radio itself can be placed where reception is the best. It will then create a WiFi network, or connect to your existing WiFi network, and provide you with a user interface accessed through a web browser on your mobile device. The software inside *Malahit-R1* is based on the open-source *OpenWebRX+* project, while the proprietary hardware is developed by *Malahiteam*.

THINGS YOU NEED

You will need the following items to get started with your *Malahit-R1* remote SDR receiver:

1. **A smartphone, a tablet, or a computer** that will serve as display and control unit. Any device that runs a fairly recent web browser will do.

2. **Four 18650 rechargeable batteries** that will go inside the receiver. You can use fewer batteries if you like (they are all connected in parallel), but fewer batteries will obviously hold less power.

3. **A laptop power supply** with the USB-C connector, to charge batteries. Any power supply that supports PD standard should work. Less powerful QC power supplies are also compatible, but may not be able to run the receiver and charge it at the same time.

4. **Three meters of wire** for the loop antenna. By default, the receiver is configured to use a loop antenna for the best performance in urban areas. It also supports other antennas though, including a telescopic one.

SETTING THINGS UP

Once you have the above items, **follow the steps outlined below:**

1. Carefully unscrew and remove end covers, then open the receiver case. **Install your 18650 batteries** into the battery holder. Make sure you observe the correct polarity. Inserting batteries in the opposite polarity may damage the receiver. Close the receiver case and screw back the end covers.

2. **Bend your antenna wire into a loop** and attach its bare ends to the pair of terminal posts found on the receiver. Move your receiver close to a window, with the loop antenna standing up against that window.

3. **Use power supply** to plug your receiver into a power outlet. Then press and release the POWER button and wait for 15-30 seconds for the receiver to start.

4. On your smartphone, table, or computer, look at the available WiFi networks and **connect to the network** listed below:

   ```
   network:  malahit
   ```

```
password: malahit-r1
```

5. Open a web browser of your choice and **go to the following web address**:

   ```
   http://192.168.10.1:8073/
   ```

   When the *Malahit-R1* web user interface loads, press the large arrow button, and start listening to the receiver. **You are all set!**

NEXT STEPS

Once you have your *Malahit-R1* running, please, **go through the following steps in order to finish setting up your receiver:**

1. Click on the *"Settings"* icon at the top-right corner of the web user interface. Then log into Settings by entering the following credentials:

   ```
   name:     admin
   password: malahit-r1
   ```

2. Click on *"General Settings"*, go to the bottom of the page, and **change your admin password** to something only you will know.

3. While at the *"General Settings"* page, you can also set up your receiver name, geographic location, the ITU region and country you are in, and other settings.

4. Once done with the *"General Settings"* page, click on *"Apply and Save"* to save the new password and other settings.

5. Go back to the main Settings page, click on *"WiFi Setup"*, and **enter the name and password for your local WiFi network**. You can have up to four different WiFi networks configured in the *"WiFi Setup"*.

6. While at the *"WiFi Setup"* page, you can also **change the name and password of the internal WiFi network** that gets activated when *Malahit-R1* cannot connect to any existing WiFi networks.

7. Once done with the *"WiFi Setup"* page, click on *"Apply and Save"* to save the new WiFi setup.

8. Click on the *"Help"* icon at the top of the web user interface to read the rest of the documentation and find out what else you can do with your *Malahit-R1* SDR receiver!

UPDATING SOFTWARE

Malahit-R1 runs Debian Linux and thus can be updated as any other Debian Linux device, by using *APT*. In order to update *Malahit-R1* software, follow these steps:

1. Connect your receiver to a local network using WiFi.

2. From another computer on the same local network, log into your receiver using SSH. Use the following credentials:

   ```
   name:     malahit
   ```

```
password: malahit-r1
```

3. Once logged in, make sure you **change the password** to something only you will know by typing this command:

   ```
   passwd
   ```

4. Obtain the list of the latest software packages by typing this command:

   ```
   sudo apt update
   ```

5. Upgrade your software to the latest versions by typing this command:

   ```
   sudo apt upgrade
   ```

6. Disconnect from the receiver by typing this command:

   ```
   exit
   ```

7. **If things go horribly wrong**, for whatever reason, you can always download a fresh SD card image from this web site:

   https://github.com/luarvique/SoapyMalahitRR/releases/

   Rewriting your SD card with a freshly downloaded image will reset Malahit-R1 to the factory settings.

CHOOSING ANTENNA

Malahit-R1 software comes configured to be used with a plain wire loop connected to the terminal posts on the receiver. It can be configured to be used with a variety of other antennas, as described below.

TELESCOPIC ANTENNAS

A short telescopic antenna can be plugged directly into the SMA connector on top of the receiver. To make *Malahit-R1* use this telescopic antenna, go to the *"Settings | SDR Device Settings | Malahit"* page and do the following:

1. Enter "Antenna" into the *"Antenna"* input field.

2. Enable the *"High-impedance antenna input"* option. Since short telescopic antennas are generally not resonant at HF frequencies, this option will help at HF. It will get automatically disabled at VHF and UHF bands.

3. Enable the *"Low-noise amplifier (LNA)"* option if you need to boost signal further. Disable this option if you are experiencing wideband distortions due to overload.

RESONANT ANTENNAS

If you are using a long, resonant HF antenna or a resonant VHF / UHF antenna, connect it to the SMA connector on top of the receiver, then go to the *"Settings | SDR Device Settings | Malahit"* page and do the following:

1. Enter "Antenna" into the *"Antenna"* input field.

2. Disable the *"High-impedance antenna input"* option.

3. Only enable the *"Low-noise amplifier (LNA)"* option if you are absolutely sure you need more boost. Most resonant antennas should not require this option.

ACTIVE ANTENNAS

If you are using an active antenna, such as active loop or miniwhip, connect it to the SMA connector on top of the receiver, then go to the *"Settings | SDR Device Settings | Malahit"* page and do the following:

1. Enter "Antenna" into the *"Antenna"* input field.

2. Disable the *"High-impedance antenna input"* option.

3. Only enable the *"Low-noise amplifier (LNA)"* option if you are absolutely sure you need more boost. Since active antennas have their own LNAs, they should not require this option.

4. If your antenna requires external Bias-T power source, enable the *"Bias-Tee power supply"* option. If your antenna comes with its own power source, make sure this option is disabled.

LOOP ANTENNAS

If your loop antenna has its own balun, connect it to the SMA connector on top of your receiver and treat it as a regular antenna. If you do not have a balun, connect your loop wire to the terminal posts on the receiver, then go to the *"Settings | SDR Device Settings | Malahit"* page and do the following:

1. For plain wires, enter "Loop" into the *"Antenna"* input field. If your loop already has its own balun, enter "Antenna".

2. Disable the *"High-impedance antenna input"* option.

3. Enable the *"Low-noise amplifier (LNA)"* option, since passive magnetic loops always require amplification.

DIPOLE ANTENNAS

In order to use a dipole, connect its shoulders to the terminal posts on the receiver, then go to the *"Settings | SDR Device Settings | Malahit"* page and do the following:

1. Enter "Loop" into the *"Antenna"* input field.

2. Disable the *"High-impedance antenna input"* option.

3. Only enable the *"Low-noise amplifier (LNA)"* option if your dipole is too short for the chosen band and thus needs more boost.
